

Nathan Fan

SKILLS

Programming Languages: Python, Java, C#, C++, JavaScript, HTML5, CSS3, SQL

Frameworks & Technologies: .NET Framework, React.js, Node.js, REST APIs, JSON, GeoJSON

Databases: MySQL, Database Design, Query Optimization, ETL Processes, Schema Design, Data Migration

Cloud & DevOps: AWS, Azure, Docker, Vercel, CI/CD Pipelines

Methodologies: Agile, Scrum, BDD, Git, Robotic Process Automation (RPA), API Integration

EDUCATION

University of Colorado, Boulder | 2025 - 2026 (expected)

Master of Science in Computer Science

GPA: 3.97 / 4.00

Algonquin College | 2020 - 2023

Associate's Degree in Computer Engineering

GPA: 3.91 / 4.00

WORK EXPERIENCE

Software Developer

Shared Services Canada | April 2024 – Present

- Built and maintained backend services and endpoints handling high volume of requests daily with 99%+ uptime
- Developed and updated RESTful APIs using C# and .NET framework for data retrieval and reporting
- Enhanced database performance by 15% through query optimization and secure patching for large-scale distributed systems
- Coordinated with cross-functional teams to deliver high-availability backend services for federal applications

Automation Developer Intern

Canada Border Services Agency | Sept 2022 – Dec 2022

- Developed Python bots to automate repetitive form processing, reducing manual workload by 30% and saving 50+ hours per month.
- Designed automation to validate and interact with distributed databases, boosting data accuracy by 22%.
- Optimized existing automation pipelines for reliability and scalability in large backend systems.
- Authored detailed user documentation and trained staff, demonstrating clear communication and collaboration.

Backend Developer Intern

RideShark | May 2022 – Aug 2022

- Built and deployed backend features in Java and Python, including robust cancellation workflows that reduced customer complaints by 35%.
- Developed GeoJSON utilities for geospatial data, speeding up distributed data processing by 20%.
- Created a transaction dashboard to monitor 1000+ financial records, improving transparency for stakeholders.
- Worked closely with frontend engineers to deliver 10+ integrated features under Agile sprints.

PROJECTS

AI Video Summarizer: An open-source application using Python, Google AI, and video processing tools to summarize video content with transcript extraction and text generation. Developed using React and NodeJS and deployed on Vercel.